文档

1. NVIDIA Container Toolkit源码：https://github.com/NVIDIA/nvidia-container-toolkit
2. NVIDIA Container Toolkit 安装向导：https://docs.nvidia.com/datacenter/cloud-native/container-toolkit/latest/install-guide.html

安装NVIDIA容器工具

确保主机操作系统已安装NVIDIA的显卡驱动

```
(GPTSoVits) nick@ubuntu:~/GPT-SoVITS4$ nvidia-smi
Sat May 18 13:48:25 2024
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 535.161.08             Driver Version: 535.161.08   CUDA Version: 12.2     |
|-----------------------------------------+----------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |         Memory-Usage | GPU-Util  Compute M. |
|                                         |                      |               MIG M. |
|=========================================+======================+======================|
|   0  Tesla P40                      Off | 00000000:84:00.0 Off |                    0 |
| N/A   36C    P0              47W / 250W |      0MiB / 23040MiB |      1%      Default |
|                                         |                      |                  N/A |
+-----------------------------------------+----------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI        PID   Type   Process name                            GPU Memory |
|        ID   ID                                                             Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

安装NVIDIA容器工具

配置apt repository（仓库）

```
curl -fsSL https://nvidia.github.io/libnvidia-container/gpgkey | sudo gpg --dearmor -o /usr/share/keyrings/nvidia-container-toolkit-keyring.gpg \
  && curl -s -L https://nvidia.github.io/libnvidia-container/stable/deb/nvidia-container-toolkit.list | \
    sed 's#deb https://#deb [signed-by=/usr/share/keyrings/nvidia-container-toolkit-keyring.gpg] https://#g' | \
    sudo tee /etc/apt/sources.list.d/nvidia-container-toolkit.list
```

安装容器工具

```
sudo apt-get update
sudo apt-get install -y nvidia-container-toolkit
```

配置

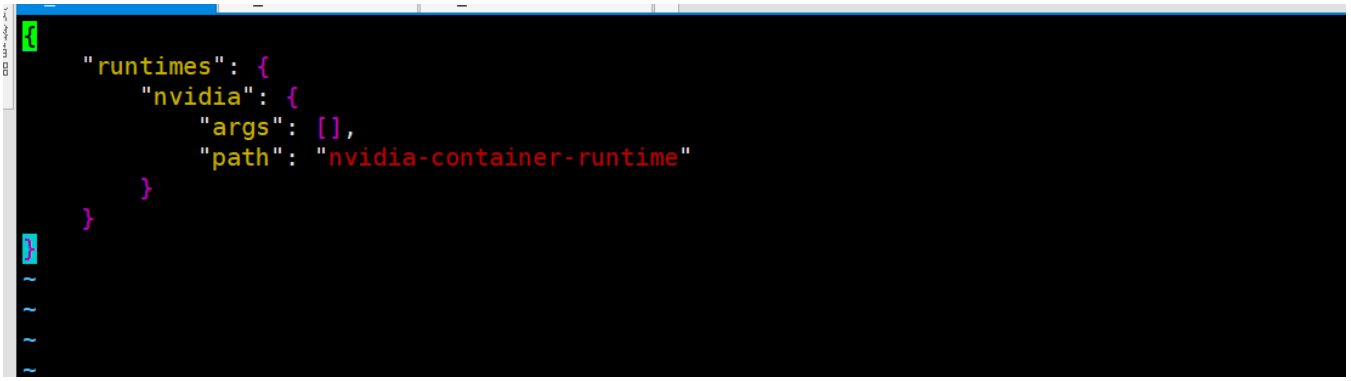
前置条件

1. 安装了支持的容器引擎（Docker、Containerd、CRI-O、Podman）
2. 安装了NVIDIA容器工具包

配置docker

1. 配置容器运行时。nvidia-ctk命令修改主机上的/etc/docker/daemon.json文件。文件会更新，以便Docker可以使用NVIDIA Container Runtime

```
sudo nvidia-ctk runtime configure --runtime=docker
```



```
{
    "runtimes": {
        "nvidia": {
            "args": [],
            "path": "nvidia-container-runtime"
        }
    }
}
~
~
~
~
```

2. 重启守护进程

```
sudo systemctl restart docker
```

运行案例

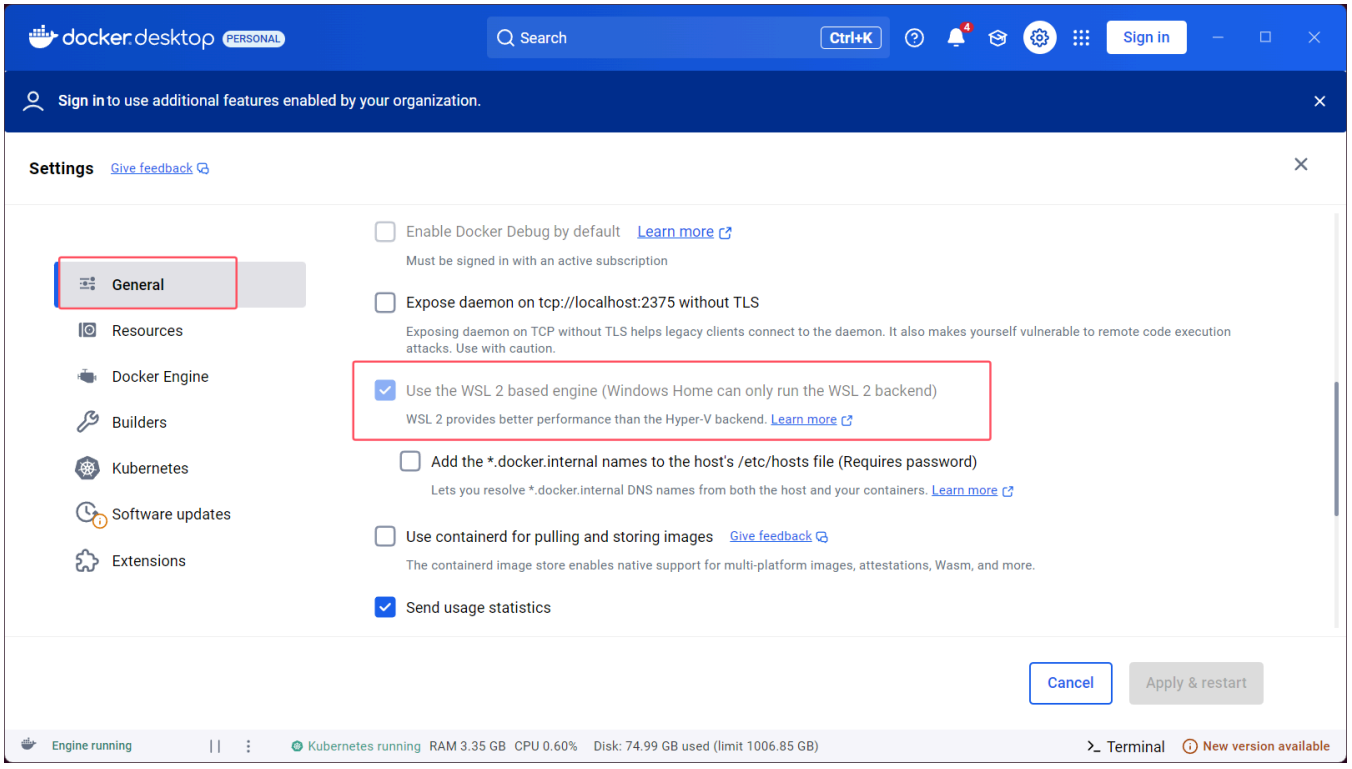
```
docker run --rm --runtime=nvidia --gpus all ubuntu nvidia-smi
```

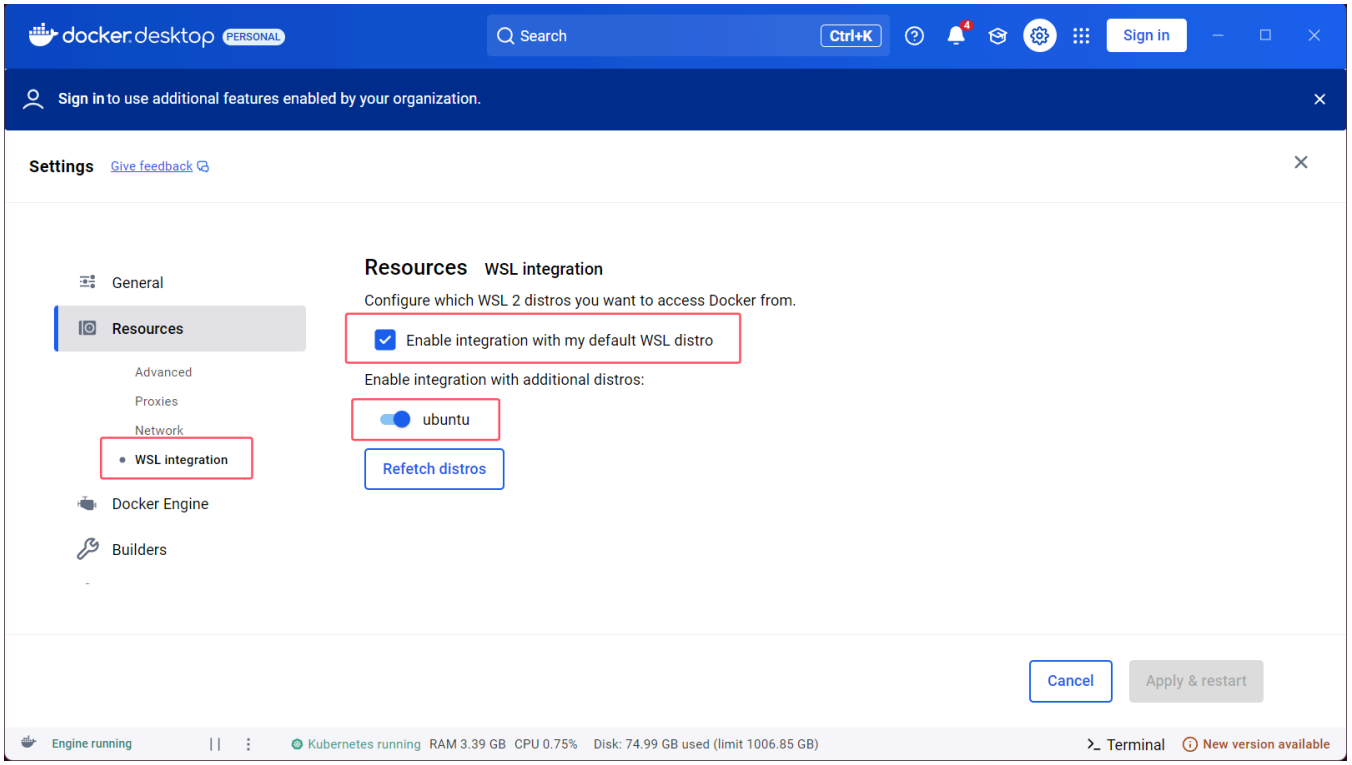
```
(base) nick@ubuntu:~$ docker run --rm --runtime=nvidia --gpus all ubuntu nvidia-smi
Mon May 20 14:10:57 2024
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 535.161.08             Driver Version: 535.161.08   CUDA Version: 12.2     |
|-----------------------------------------+----------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |         Memory-Usage | GPU-Util  Compute M. |
|                                         |                      |               MIG M. |
|=========================================+======================+======================|
|   0  Tesla P40                      Off | 00000000:84:00.0 Off |                    0 |
| N/A   31C    P0              46W / 250W |      0MiB / 23040MiB |      1%      Default |
|                                         |                      |                  N/A |
+-----------------------------------------+----------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI        PID   Type   Process name                                  GPU Memory |
|        ID   ID                                                                   Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

windows docker.desktop 设置

docker.desktop 设置




docker desktop PERSONAL
Search
Ctrl+K
Sign in
Sign in to use additional features enabled by your organization.
Settings
Give feedback
General
Resources
Advanced
Proxies
Network
WSL integration
Docker Engine
Builders
Resources WSL integration
Configure which WSL 2 distros you want to access Docker from.
Enable integration with my default WSL distro
Enable integration with additional distros:
ubuntu
Refetch distros
Cancel
Apply & restart
Engine running
Kubernetes running RAM 3.39 GB CPU 0.75% Disk: 74.99 GB used (limit 1006.85 GB)
Terminal
New version available